

LeetCode Notes: Remaining Methods

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Logic & Approach

Initial Observation: Should be marked easy.

Step 1: Marking Suspicious Nodes

Basic aadya ye rhega ki `k` se start krte hue bfs ya dfs kr lenge aur saare sus nodes mark kr lege.

Step 2: Checking Connections

Ab agar kisika source good he aur destination sus he to matlab uska `goodindegree > 0` he. Hence cant remove. Else remove the whole group.

Step 3: Final Decision

Saare khraab nodes mark kr liye. Ab dekho ki kya koi khraab node kisi acche node se connected he:

- Agar ha, to removal is impossible (saare node return kr do).
- Agar na, to sab khraab ko remove kr do (sirf good nodes wapas krenge).

C++ Solution

```
class Solution {
public:
    vector<int> remainingMethods(int n, int k, vector<vector<int>>& invocations)
    {
        //should be marked easy
        //basic aadya ye rhega ki
        //k se start krte hue bfs ya dfs kr lenge aur saare sus nodes mark kr
        lege
        //ab agar kisika source good he aur destination sus he
        //to matlab uska goodindegree>0 he
        //hence cant remove
    }
};
```

```

//else remove the whole group

vector<vector<int>> adj(n);
for(auto &it : invocations) {
    int u = it[0];
    int v = it[1];
    adj[u].push_back(v);
}

queue<int> q;
vector<int> sus(n, 0);

q.push(k);
sus[k] = 1;

while(!q.empty()) {
    int curr = q.front();
    q.pop();

    for(auto &it : adj[curr]) {
        if(sus[it]) continue;
        sus[it] = 1;
        q.push(it);
    }
}

//saare khraab nodes mark kr liye
//ab dekho ki kya koi khraab node kisi acche node se connected he
//agar ha , to removal is impossible
//agar na , to sab khraab ko remove kr do

vector<int> goodindeg(n, 0);
for(auto &it : invocations) {
    int u = it[0];
    int v = it[1];

    if(sus[u] == 0 && sus[v] == 1) {
        goodindeg[v]++;
    }
}

for(int i = 0; i < n; i++) {
    if(sus[i] == 1 && goodindeg[i] > 0) {
        //matlab impossible to remove saare node return kr do
        vector<int> ans;
        for(int j = 0; j < n; j++) {
            ans.push_back(j);
        }
        return ans;
    }
}

//agar waisa nhi hua to removable he , sirf good nodes wapas krenge
vector<int> ans;

```

```
    for(int i = 0; i < n; i++) {  
        if(sus[i] == 0) {  
            ans.push_back(i);  
        }  
    }  
  
    return ans;  
}  
};
```

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